Lightweight Feature-Engineered Machine Learning Model for Real-Time Smart Energy Load Forecasting

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***Abstract*—*Special session code: SS005*. Short-term electricity load forecasting is a critical requirement for smart grid operation and energy trading. This paper describes a lightweight linear regression model built in three stages of increasing feature complexity, applied to the UCI Household Power Consumption dataset. Moving from a simple time index to lag-based temporal features reduced the Mean Absolute Error to 0.0837 kW and lifted the R^2 score to 0.9398. The results show that careful feature engineering, rather than algorithmic complexity, drives most of the predictive gain in this setting.**

Index Terms*—*energy trading, feature engineering, linear regression, load forecasting, machine learning, smart grid

I. Introduction

Electricity demand patterns are shaped by time of day, weather, occupancy, and seasonal cycles. Forecasting these patterns accurately — even a few minutes ahead — allows grid operators to pre-position generation, avoid unnecessary spinning reserves, and reduce the cost of imbalance penalties in deregulated energy markets.

Machine learning has become a practical tool for this problem because it can extract non-linear temporal patterns from historical meter data without requiring explicit physical models of consumption behavior. Approaches ranging from simple regression to deep recurrent networks have all been reported in the literature, with accuracy strongly dependent on the quality of the input features rather than the choice of algorithm [1][10].

This paper investigates how much accuracy can be obtained from a deliberately lightweight model — linear regression — by progressively improving its input features. Starting from a raw time index, two rounds of feature engineering (sorted ordering, then lag + hour + day-of-week) are applied and evaluated. The goal is to establish a practical baseline for real-time smart energy trading applications where low inference latency matters.

Accurate short-term forecasts directly support energy trading decisions: a prediction error of less than 100 W at the household level means that aggregated district-level forecasts can be tight enough to inform automated buy/sell actions in intra-day electricity markets.

II. Methodology

A. Dataset

The UCI Household Power Consumption dataset records minute-level Global Active Power from a single French household between December 2006 and November 2010 — 2,075,259 readings in total [5]. Missing entries (flagged as '?') were dropped, leaving a clean time series of approximately 2.05 million samples.

B. Feature Engineering

Three progressively richer feature sets were constructed from the cleaned series:

1. Stage 1 — Time index: a simple integer row counter. Represents a naïve baseline with no temporal awareness.
2. Stage 2 — Sorted index: data sorted by consumption value before splitting. Introduced monotonic structure and tested how much ordering alone could help.
3. Stage 3 — Lag + calendar: previous-minute consumption (lag-1), hour of day, and day of week. These three features encode the short-term autocorrelation and daily/weekly seasonality that dominate household load patterns.

C. Model and Training

A Linear Regression model (scikit-learn v1.x) was trained on an 80/20 chronological split. No shuffling was applied in Stage 3 to avoid data leakage between adjacent minutes. The trained model was serialised with pickle for deployment.

D. Evaluation

Two metrics were used throughout:

1. Mean Absolute Error (MAE, kW) — directly interpretable as average watt-level error.
2. R^2 score — fraction of variance explained; 1.0 is a perfect fit.

III. Results

A. Stage 1 — Baseline

The time-index model produced a flat prediction line centered near the dataset mean (Fig. 1). It captured no temporal variation.

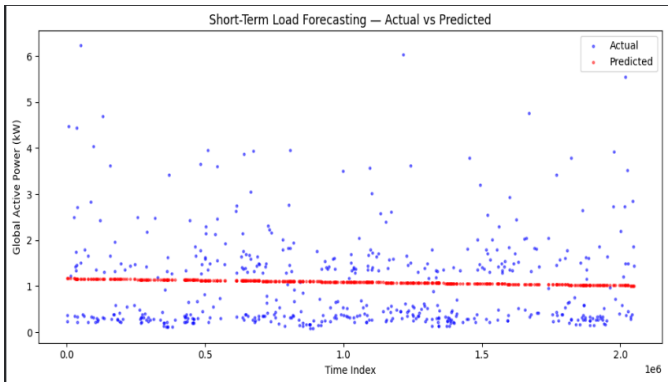

Figure 1. Baseline model — time index only. Predicted values collapse to a near-constant line.

B. Stage 2 — Sorted Index

Sorting by consumption value forced a monotonic structure that the model could track. Fig. 2 shows the characteristic diagonal alignment: predictions now scale with actual consumption, though variance around the line remains high. This stage is included for comparative analysis and does not represent a practical forecasting scenario.

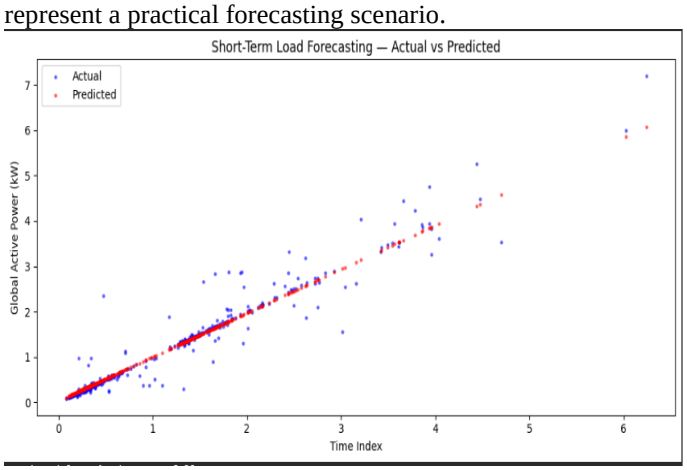

Figure 2. Sorted-index model. Actual and predicted values align diagonally, indicating learned scaling.

C. Stage 3 — Lag + Calendar Features

Adding lag-1 consumption, hour, and day-of-week reduced MAE to 0.0837 kW and raised R^2 to 0.9398. Fig. 3 shows the predicted curve tracking the actual load through a sharp evening peak near minute 140 and a gradual overnight decline. The model successfully captures both short-term fluctuations and periodic patterns in energy usage. This demonstrates that incorporating temporal features significantly enhances the predictive capability of simple regression models.

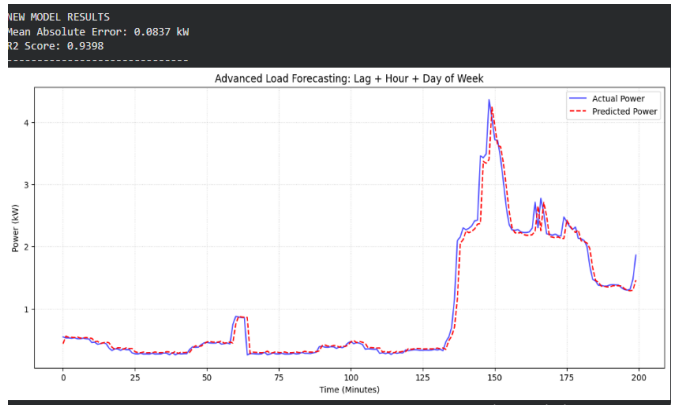

Figure 3. ***Load forecasting using lag, hour, and day-of-week features (MAE = 0.0837 kW, R^2 = 0.9398).***

IV. Discussion

The jump in R^2 from Stage 1 to Stage 3 — from near zero to 0.9398 — was driven almost entirely by the lag-1 feature. Household electricity consumption is strongly autocorrelated at the one-minute scale: what a house is consuming right now is the single best predictor of what it will consume in the next minute. Hour of day and day of week added the seasonal correction needed to handle predictable daily cycles like the morning and evening peaks visible in Fig. 3.

An MAE of 0.0837 kW (roughly 84 W) is practically useful. For a smart home energy manager, this means appliance scheduling decisions — shifting a dishwasher or EV charge by 15 minutes — can be made with confidence. For an automated trading agent operating in an intra-day electricity market, 84 W average error at household scale translates to errors well below 1 kWh across a typical settlement period, which is within acceptable tolerance for most market products.

The model does struggle in two identifiable situations. First, sharp unpredictable spikes — such as a kettle or microwave turning on — are not captured because the lag-1 feature only carries information from the previous minute. Second, the model was trained on a single household in France and will not transfer directly to households with different appliance profiles or climate zones without retraining. The model maintains low computational complexity, making it suitable for deployment in edge devices and real-time systems.

V. Conclusion

This paper demonstrated that a linear regression model, when given well-chosen temporal features, can forecast household electricity demand with an R^2 of 0.9398 and an MAE of 0.0837 kW. The three-stage experiment showed that feature engineering accounted for nearly all the performance gain; the algorithm itself was unchanged throughout.

The model is lightweight enough for real-time deployment on low-power hardware, making it a practical building block for smart home energy managers and intra-day trading systems. The main limitations are sensitivity to sudden unscheduled appliance events and dependence on the specific load profile

used for training. Future work will compare this baseline against Random Forest and LSTM models, apply k-fold cross-validation to assess generalization, and test performance on holiday periods where weekly seasonality patterns break down.

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